

Module 8

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Module 8

Exception Handling

In Module 8 we tackle the problem of error handling. Error handling is hard no matter what language you're programming in. There are two issues:

1. How do you pass control from the place where the error occurred (the error *site*), to a place where you can deal with the error (the error *handler*)?
2. And, once you're there, what kind of corrective actions can you take?

We consider only the first issue in any detail, since the second has a mind-boggling assortment of answers, based on the problem you're solving. In the commentary, we first consider a language independent scheme for handling errors and discuss its inadequacies. Then we look at the C approach, and again see why it is inadequate for C++. Finally, we look at the C++ exception mechanism for transferring control from the error site to the error handler.

[Learning Objectives](#)

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Learning Objectives

- Error handling is hard: the general problem
- A language-independent solution
- The C solution: `setjmp` & `longjmp`
- The C++ solution: exceptions
- Try and catch blocks
- Exception hierarchies and passing data in exceptions

Lectures

Presentation

- **[8.1: Comments on Reading Assignments for Module 7](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-8-dot-1>)
- **[8.2: Error Handling](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-8-dot-2>)
- **[8.3: Exceptions](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-8-dot-3>)
- **[8.4: Exception Hierarchies and Passing Data](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-8-dot-4>)
- **[8.5: More about Exceptions](#)**
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-8-dot-5>)

Reading

The reading for this Module is C review. Depending on your C background, you may be able to move through this material quickly.

Required Readings

- *C++ Primer Plus*, Chapter 15

Recommended Readings

- *Thinking in C++ Volume One*, pp. 681–720

Discussion

[Module 8 Discussion](#)

https://onlinelearning.berkeley.edu/courses/2291660/discussion_topics/14351906

Assignment

[Module 8: Short Answer 8](#)

<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623166>

[Module 8: Programming Assignment 8](#)

<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623164>

Source Code

In each module of this course, you will be required to submit a programming assignment. For each module, I have prepared sample source code for your reference. Please take a few minutes to download and install all of these programming examples. The files have been compressed into one zip. You will need a zip utility to decompress the file.

 [sources.zip](#)

<https://onlinelearning.berkeley.edu/courses/2291660/files/313890817/download?wrap=1>